

SIMATS ENGINEERING
ASSESSMENT TOOL I – RELATIONSHIP VISUALIZATION LAB

Course: Data Handling and Visualization	Code: DSA0603	CO: CO4
Duration: 60 min	Total Marks: 50	Reg No : 192324285 Name: Surya JK

COURSE OUTCOME COVERED

CO1 Understand the fundamental concepts, principles, and techniques of data visualization. (BL3)

Activity Tool : Relationship Visualization Lab

Weightage: 20%

Code of Conduct:

I Surya JK / 192324285 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Selection of Relationship Visualization 25%	Relationship Analysis 25%	Application of Visualization Techniques 20%	Organization 15%	Accuracy 10%	Clarity 5%	Total
Q1							
Q2							
Q3							
Q4							
Q5							

RELATIONSHIP VISUALIZATION LAB

QUESTIONS WITH ANSWERS

Questions and answers based on the supplied Assessment Tool I. Question 3 is only partially specified in the source, and the actual network_edges.csv values are not included in the supplied PDF, so those portions are explicitly identified rather than invented.

Question 1: Scatterplot with Regression Diagnostics

Question: Using mtcars.csv, create a scatterplot of mpg vs wt using ggplot2. Add a linear regression line with confidence interval, color points by cyl as a factor, size points by hp, calculate the Pearson correlation coefficient between mpg and wt, and interpret the direction and strength.

Answer: The Pearson correlation between mpg and wt is approximately -0.868. This indicates a strong negative relationship: as vehicle weight increases, fuel economy (mpg) tends to decrease. The relationship is strong because the absolute correlation value is close to 1.

R Code:

```
library(ggplot2)
data(mtcars)
ggplot(mtcars, aes(x = wt, y = mpg, color = factor(cyl), size = hp)) +
  geom_point() +
  geom_smooth(method = "lm", se = TRUE) +
  labs(x = "Weight (wt)", y = "Miles per Gallon (mpg)",
       color = "Cylinders", size = "Horsepower",
       title = "MPG vs Vehicle Weight")
cor(mtcars$mpg, mtcars$wt, method = "pearson")
```

Question 2: Correlogram Construction

Question: Using mtcars and all numeric columns, compute the correlation matrix and visualize it as a correlogram using corrplot or GGally::ggcorr. Display numeric correlation coefficients, use a blue–white–red diverging scale, reorder variables using hierarchical clustering, and identify the strongest positive and negative variable pairs.

Answer: The strongest positive correlation is between wt and disp (approximately +0.888). The strongest negative correlation is between mpg and wt (approximately -0.868). The correlogram should use numeric labels, a blue–white–red diverging scale, and hierarchical clustering order.

R Code:

```
library(corrplot)
data(mtcars)
cor_mat <- cor(mtcars)
corrplot(cor_mat, method = "color", type = "upper",
         order = "hclust", addCoef.col = "black",
         tl.col = "black",
         col = colorRampPalette(c("blue", "white", "red"))(200))
round(cor_mat, 3)
```

Question 3: Dimension Reduction Comparison (PCA vs. t-SNE)

Question: The supplied document provides the heading “Dimension Reduction Comparison (PCA vs. t-SNE)” on page 3, but it does not include the detailed dataset, plotting requirements, or exact sub-questions for Question 3.

Answer: A source-specific answer cannot be determined from the supplied document because the detailed instructions for Q3 are missing. In general, PCA is a linear dimensional-reduction method that transforms correlated variables into orthogonal principal components while preserving as much variance as possible. t-SNE is a nonlinear technique mainly used to visualize high-dimensional data in two or three dimensions by preserving local neighborhood structure. PCA is generally faster and more interpretable, while t-SNE can reveal nonlinear clusters but depends on parameters such as perplexity and random initialization.

Question 4: Bubble Chart for Multivariate Paired Data

Question: Using gapminder2007_practice.csv (or the gapminder package with year == 2007), filter for 2007 and create a bubble chart with gdpPercap on a log-transformed x-axis, lifeExp on the y-axis, pop as bubble size, and continent as bubble color. Add suitable labels/title, control bubble size using scale_size_continuous(), and discuss one visible cluster or outlier.

Answer: The chart should generally show that countries with higher GDP per capita tend to have higher life expectancy, although the relationship is not perfectly linear. Very large population bubbles can stand out in size, while countries with unusually high or low life

expectancy relative to GDP per capita can stand out as relationship outliers. Because the supplied dataset is described as synthetic, the exact visible cluster or outlier should be identified from the generated plot.

R Code:

```
library(ggplot2)
gm <- read.csv("gapminder2007_practice.csv")
gm2007 <- subset(gm, year == 2007)
ggplot(gm2007, aes(x = gdpPercap, y = lifeExp,
                  size = pop, color = continent)) +
  geom_point(alpha = 0.7) +
  scale_x_log10() +
  scale_size_continuous(range = c(2, 15)) +
  labs(x = "GDP per Capita (log scale)", y = "Life Expectancy",
       size = "Population", color = "Continent",
       title = "Life Expectancy vs GDP per Capita, 2007")
```

Question 5: Hexbin Plot vs. Network Graph

Question: Part (a): Simulate two correlated variables with 10,000 observations and create a hexbin plot using `geom_hex()`. Explain why it is preferable to a standard scatterplot. Part (b): Using `network_edges.csv`, build a network graph using `igraph` or `ggraph`, size nodes by degree centrality, color edges by weight, report the node with the highest centrality, and explain its role.

Answer: Part (a): With 10,000 observations, a standard scatterplot can suffer from severe overplotting. A hexbin plot groups observations into hexagonal bins and represents the number of observations in each bin, making dense regions and the overall relationship easier to see. Part (b): The supplied PDF does not contain the actual edge-list values from `network_edges.csv`, so the exact highest-centrality node cannot be determined from the PDF alone. After loading the CSV, calculate degree centrality and report the node with the largest value. A highly central node is strongly connected to many other nodes and may act as a hub or important connector.

R Code:

```
# Part (a)
library(ggplot2)
set.seed(123)
x <- rnorm(10000)
y <- 0.7 * x + rnorm(10000, sd = 0.7)
sim_data <- data.frame(x = x, y = y)
ggplot(sim_data, aes(x = x, y = y)) +
  geom_hex() +
  labs(title = "Hexbin Plot of Two Correlated Variables",
       x = "X", y = "Y")

# Part (b)
library(igraph)
edges <- read.csv("network_edges.csv")
g <- graph_from_data_frame(edges, directed = FALSE)
deg <- degree(g, mode = "all")
centrality <- data.frame(node = names(deg), degree = as.numeric(deg))
centrality <- centrality[order(-centrality$degree), ]
centrality
plot(g, vertex.size = 5 + 5 * deg,
     edge.width = 1 + E(g)$weight)
```